Computational Materials Science 249 (2025) 113651

Contents lists available at ScienceDirect

Computational Materials Science

journal homepage: www.elsevier.com/locate/commatsci

Full Length Article

Machine learning research advances in energy storage polymer-based dielectrics

Qixin Yuan [a,b], Dong Yue [a,b,*], Zhe Zhang [a,b], Yu Feng [a,b,*], Qingguo Chen [a,b,*]

[a] Key Laboratory of Engineering Dielectrics and its Application, Ministry of Education. Harbin University of Science and Technology, Harbin 150080, PR China
[b] School of Electrical and Electronic Engineering, Harbin University of Science and Technology, Harbin 150080, PR China

ARTICLE INFO

Keywords:
Polymer-based dielectrics
Machine learning
Energy storage
Performance prediction

ABSTRACT

In the new circumstances of modern scientific research combining advanced analytics and artificial intelligence, the application of machine learning (ML) to energy storage dielectrics has become the focus of research attention in this field. In this review, the current disciplinary fields and basic workflow of ML applications are summarized and the important impact of ML in energy storage polymer-based dielectric research is emphasized, with a focus on enabling rapid performance prediction and accelerating the research and development of novel materials. The content focuses on several common methods and representative algorithms for establishing databases, including dataset collection results, model calculation results, and experimental verification results. Moreover, the advantages and disadvantages of each method of dataset collection and the accuracy and reliability of each algorithm prediction application are summarized and compared. Finally, based on ML's impact on the research field of energy storage polymer, its prospects and challenges are discussed. This review not only provides the latest progress of existing researchers in using ML in energy storage polymers but also looks forward to providing new modes for the preparation of high-energy storage polymer-based dielectrics through ML.

1. Introduction

As global population and energy consumption rise, human society faces climate change, resource depletion, and ecological issues [1–3]. Renewable energy can mitigate these challenges but suffers from intermittency and volatility, hindering its widespread application. Improving energy storage technology is crucial to achieving a green, sustainable energy supply [4,5]. Research focuses on electrostatic capacitors (dielectric capacitors), electrochemical capacitors, batteries, and solid oxide fuel cells, as shown in Fig. 1 [6]. Dielectric capacitors, with ultra-fast charge–discharge rates and high energy density, show significant potential in renewable energy and electric vehicles [7–9].

Polymer-based dielectrics are the key materials of dielectric capacitors [10–12], which have high voltage resistance, low loss, high flexibility, and good cost-effectiveness [13,14]. Frequently utilized polymers include biaxially oriented polypropylene (BOPP) [15], polypropylene (PP) [16], polyimide (PI) [17], polyetherimide (PEI) [18], polycarbonate (PC) [19], polyethylene terephthalate (PET) [20], poly (methyl methacrylate) (PMMA) [21], fluorene polyester (FPE) [22], polyvinylidene fluoride (PVDF) [23], and others. However, with the growing demand for high efficiency and miniaturization, the low energy storage density of dielectric capacitors has become a barrier to their further development [24]. Therefore, the development of dielectric materials with high energy storage density is extremely urgent. Fig. 2 illustrates the breakdown and energy storage properties of widely recognized polymers.

The storage and release of electrical energy in a dielectric capacitor is attributed to the polarization, and its structure is exhibited in Fig. 3(a), which is composed of two parallel electrode plates [25,26]. The polarization behavior can be calculated from the *P-E* loop [27], as exhibited in Fig. 3(b). The energy storage density U_e calculation formula is as exhibited in Eq. (1):

$$U_e = \int_{P_r}^{P_{max}} EdP \tag{1}$$

where E is the applied electric field, P_{max} is the maximum polarization, and P_r is the residual polarization. The red region in Fig. 3(b) equates to the discharged energy storage density U_e; the blue region equates to the dissipated energy storage density U_{loss}. The macroscopic parametric relationship between the polarization intensity (P) and the permittivity (ε_r) and E is exhibited in Eq. (2):

* Corresponding authors at: Harbin University of Science and Technology, Harbin 150080, PR China.
E-mail addresses: yuedong@hrbust.edu.cn (D. Yue), fengyu@hrbust.edu.cn (Y. Feng), qgchen@263.net (Q. Chen).

https://doi.org/10.1016/j.commatsci.2024.113651
Received 20 August 2024; Received in revised form 18 December 2024; Accepted 27 December 2024
Available online 11 January 2025
0927-0256/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

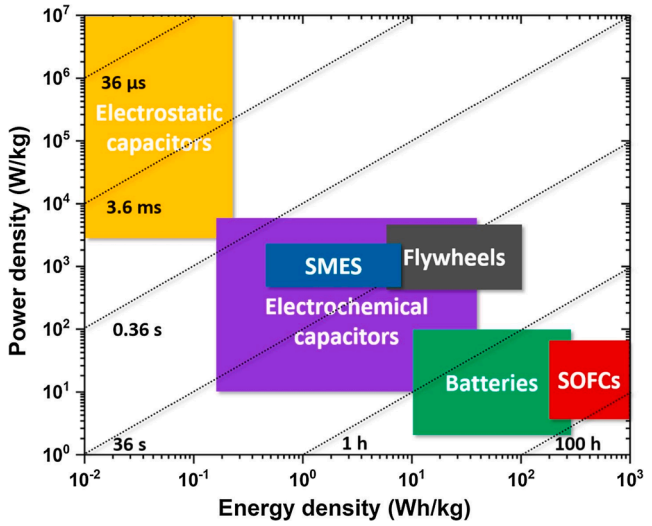


Fig. 1. Characteristics of different energy storage devices. Reproduced with permission. [6] Copyright 2019, Elsevier.

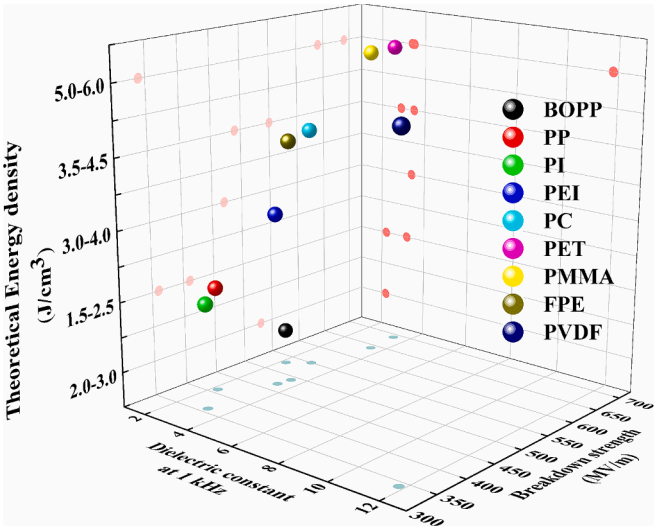


Fig. 2. Performance Parameters of Common Energy Storage Polymers.

$$P = \varepsilon_0(\varepsilon_r - 1)E \tag{2}$$

where ε_0 is the vacuum permittivity (8.85×10^{-12}F m^{-1}). According to Eqs. (1) and (2), high P_{max} or low P_r and high E_b are needed for obtaining high U_e.

According to the definition of the dielectric polarization intensity P, the relationship with microscopic parameters is exhibited in Eq. (3):

$$P = N\mu \tag{3}$$

where N is the number of molecules per unit volume of the dielectric, μ is the induced dipole moment of each molecule in the dielectric of direction electric field. If the effective electric field strength acting on the molecule is E_i, the induceddipole moment of the molecule can be considered to be proportional to E_i, as exhibited in Eq. (4):

$$\mu = \alpha E_i \tag{4}$$

The proportional constant α is the molecular polarizability, so according to Eqs. (2), (3), and (4), it can be obtained that

$$P = \varepsilon_0(\varepsilon_r - 1)E = N\alpha E_i \tag{5}$$

or

$$\varepsilon_r - 1 = \frac{N\alpha E_i}{\varepsilon_0 E} \tag{6}$$

Eq. (6) establishes a connection between ε_r and N, α, E_i. According to the microscopic mechanism of dielectric polarization, it can be summarized into four basic forms, i.e., electron displacement polarization, ion displacement polarization, dipole shift to polarization, and interface polarization [6,28,29], and the frequency range of each polarization is exhibited in Fig. 4. The frequency range of electron displacement polarization is about 10^{14} to 10^{16} Hz, resulting from the electric dipole moment formed by the relative displacement of the nucleus and the electron cloud under the electric field is exhibited in Fig. 4(a). The frequency range of ion displacement polarization is 10^{9} to 10^{13} Hz, resulting from the electric dipole moment formed by heteropolar ions moving in the reverse orientation under the electric field is exhibited in Fig. 4(b). The frequency of dipole steering polarization is about 10^{3} to 10^{8} Hz, resulting from the unbalanced division of dipoles inside atoms or ions arranged along the electric field, as exhibited in Fig. 4(c). The frequency range of interfacial polarization is 10^{-3} to 10^{2} Hz, which is caused by the inhomogeneity of the internal components of the dielectric, such as impurities and defects, and some free charges remain in the trapping center or interface, which cannot be completely neutralized, forming a space charge layer, as exhibited in Fig. 4(d).

Polymer-based dielectrics ε_r are used to describe the ability of the dielectric in electric field polarization; it depends on the matrix of ε_r and the parameters of fillers [30,31]. At present, mainly three kinds of dielectric material, i.e., ceramic materials, organic polymers, and polymer-inorganic filler composite materials, respectively [32]. It appears that a constraint between ε_r and E_b, with ceramics and polymers being the two limits [24]. As for ceramic materials, they generally have the advantages of high dielectric constants (10^{2} to 10^{4}), long cycle life, and a broad temperature and frequency range, but the high number of pores in the ceramic blocks leads to low breakdown field strengths, which greatly limits the improvement of their energy storage densities

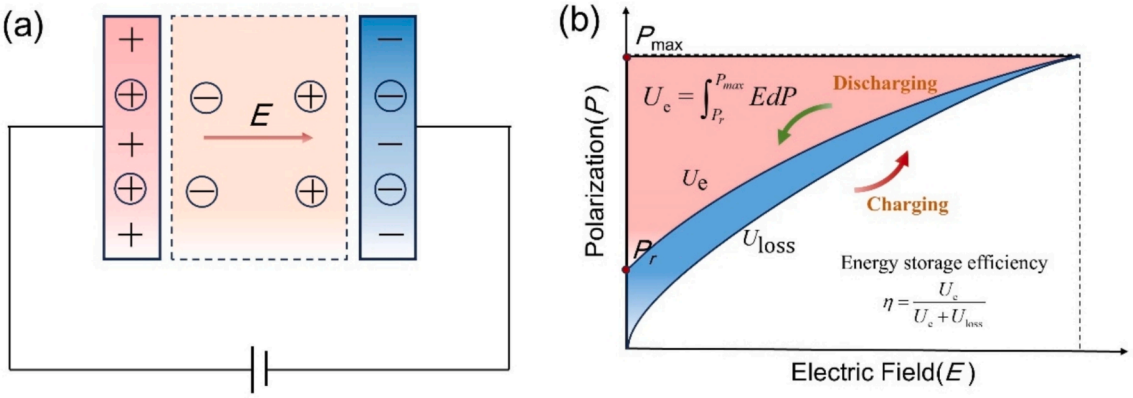


Fig. 3. (a) Charge and electric field distribution of a plate capacitor charged with a medium. (b) Polarimetric electric field (*P-E*) hysteresis loop.

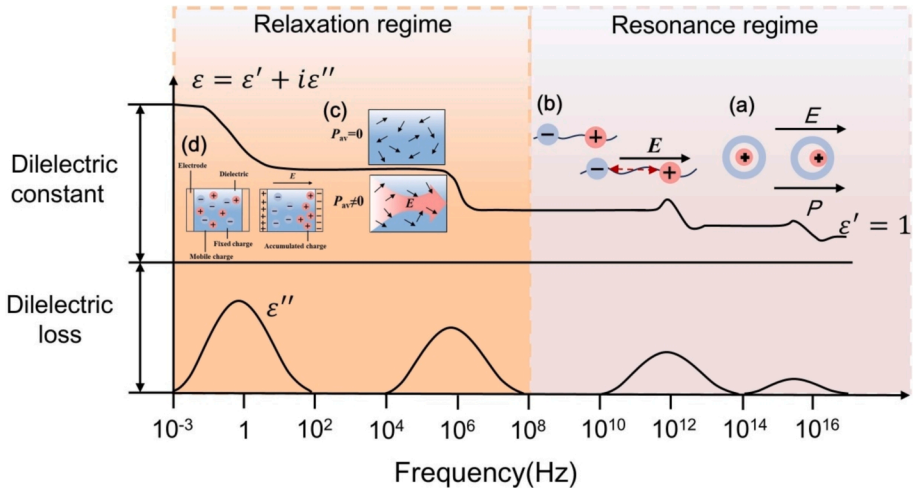


Fig. 4. The relationship between polarization and frequency. Electronic (a), ionic (b), dipole (c), and interfacial polarization (d).

[33,34]. Organic polymer materials have very high breakdown field strengths, excellent mechanical properties, and processability, but due to their low dielectric constants (1 to 10), their energy storage densities are also not high [35–37]. Attributed to the dielectric constant reflecting the ability to bind charges. The higher the dielectric constant, the easier it is for the electron to move completely, which means that under the electric field, the carrier density in the dielectric is higher, and the avalanche multiplication trigger is earlier, resulting in a lower breakdown strength [32]. Based on articles related to energy storage polymers consulted on the Web of Science, the characteristics of dielectric constant, breakdown strength, and energy density of pure polymers, bulk ceramics, polymer/polymer composites, and polymer/ceramic composites are summarized, as exhibited in Fig. 5 [32]. The energy density of the dielectric nanocomposites is greatly increased with the fusion of the polymer and inorganic phases, proving the workability of this method [32]. Therefore, to realize ε_r and E_b increase at the same time, combining the inorganic substance with organic polymers and thus obtaining excellent energy storage performance [32,38].

The determinants of energy storage density in composite include the matrix structure, the shape and size of the fillers, and the microstructure [39–41]. The traditional design of experiments involves a large number of performance experiments on several candidates that may satisfy the desired requirements and then selecting the best candidate from among them to be investigated and optimized in detail, and the drawbacks of this process are obvious, i.e., inefficiency, high cost, and long research and development period. With the rapid rise of computational

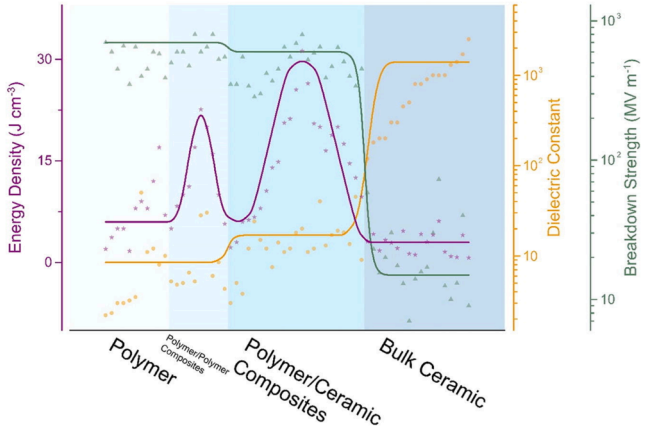


Fig. 5. Changes in performance parameters of four materials. Reproduced with permission. [32] Copyright 2019, Elsevier.

technology, computer-aided simulation stands out in materials design and performance prediction which covers wide spatial and temporal scales, e.g. micro-level simulations (e.g., first-principles (FP), calculations, quantum mechanics (QM), and molecular dynamics (MD)) [42–44], mesoscale simulations focusing on phase-field (PF) and force-field (FF) [45,46], and macro-level simulations based on finite element (FE) and finite difference (FD), which assist researchers in exploring the correlation between material properties and structures [47–49]. The performance of composite materials is linked to multidimensional factors in the synthesis and processing process, and the limited validity of experimental measurements and computational simulations reduces accuracy [50–52]. In today's rapidly developing era of informatization and digitization, artificial intelligence (AI) and big data have become the two pillars of progress [53], machine learning (ML) has been successfully applied to the processing of large amounts of data [54–56]. Over the past decade, the application of machine learning techniques in materials research has evolved from a niche topic to a full-fledged subfield in the field of materials science and engineering. Machine learning has gone through several stages of theoretical proposal, development bottleneck, technological recovery and explosive growth, and in recent years, with the improvement of computing power, big data application and algorithmic innovation, it has made leaps and bounds and rapidly expanded to various disciplines and application areas. Machine learning is at the heart of the fourth paradigm, where investigators use ML methods to analyze data and find patterns [57–59]. It typically goes through the following key steps before making a prediction or decision: data processing, feature extraction, model training, and model evaluation.

Recently, ML has been widely applied in various fields. There are already many excellent reviews on different aspects of materials machine learning, including its application in simulation and modelling [60], synthesis and characterisation [61], fabrication [62], and literature mining [63]. For example, Li et al. [64] selected the XGBoost algorithm with a higher coefficient of determination R^2 and screened borate chalcogenides with high E_b from a dataset of 760 chalcogenide oxides. Chandan R et al. [65] performed complex dielectric constant measurements at 201 frequency points for aqueous solutions of MB over a limited range of concentrations and temperatures and used the CatBoost regression model to predict the dielectric properties at intermediate frequencies and at different concentrations to reduce experimental cost and time. Chen et al. [66] utilized a dataset of 1210 experimental measurements at different frequencies and a Gaussian process regression algorithm for training to estimate the dielectric constants of 11,000 polymers dataset and proposed five representative polymers for capacitor and microelectronic applications. Therefore, ML may be a powerful

approach to find potential microstructure-property relationships, reverse-engineer polymer nanocomposites dielectric, and accelerate the prediction of material properties [67]. Using polymers as a research base, the number of publications from 2012 to 2023 is exhibited in Fig. 6, using the subject word 1 "polymers, energy storage" (corresponding to the blue graphic), and subject word 2 "polymers, energy storage, machine learning" (corresponding to the red graph), as exhibited (a) and (b). The upward trend of the bar chart indicates that ML in polymer-based energy storage has been increasing year by year over the past 12 years.

To navigate the growth challenges in the energy storage polymer-based dielectrics, this paper reviews the current state of ML in energy storage polymer research, segmenting the composition of the ML segments as exhibited in Fig. 7. The second part discusses the method of collecting data and the comparison of advantages and disadvantages. The third part discusses the accuracy of the ML output results and the reliability of laboratory-based verification of them. Finally, the current challenges faced by ML in this field and possible future developments are presented. Fig. 6 also provides statistics on the number of publications using the subject words "polymers, energy storage" and "polymers, energy storage, machine learning" from 2012 to 2023. The number of publications from 2012 to 2023 is 832 to 5,061 and 0 to 29 for subject words 1 and 2, respectively. Since 2020, the application trend of ML in research and development strategies for energy storage polymers has significantly increased. Looking forward this review can act as a favorable tool to assist researchers in creating more excellent results in the future.

2. Dataset creation

Machine learning is used as a scientific and technological tool for materials development and optimization, and the data is a key element used in the process of predictive analysis. As proposed at the Xiangshan Science Conference to launch China's MGI, the MGI in the US, and the European Union, Japan, and Russia to launch similar materials research programs, all countries share the common goal of establishing high-quality materials databases and emphasizing the breadth of the data contained therein [67,68]. The overall environment of emphasis on data certainly creates opportunities for data-driven ML technologies. Until now, four methods have been utilized for dataset collection in all, e.g., dataset collection from existing databases, literature, high throughput computing, and experimental data.

2.1. Dataset collection from public databases and platforms

Using available databases for research has been widely adopted

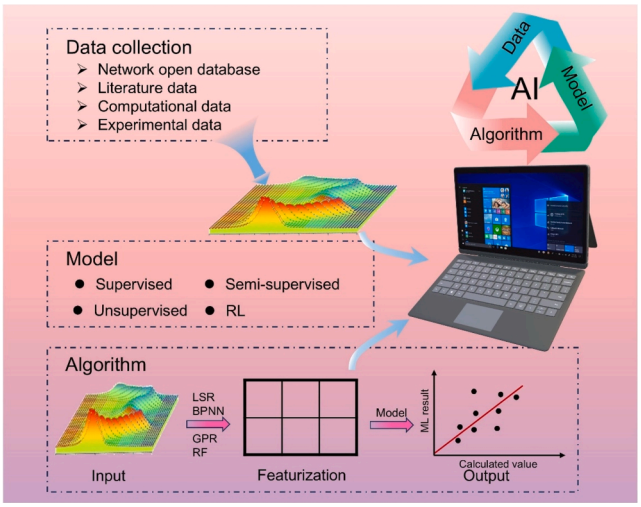


Fig. 7. Machine Learning Workflow.

[67,69,70], and databases and platforms such as Scikit-learn [71,72], PolyInfo [73], Polymer Genome [74], PI1M [75], etc. have been explored. Building highly accurate machine learning models based on data points from publicly available databases is the greatest strength of the method. However, the isolation of these databases does not allow for simultaneous multivariate cross-searching, reducing the availability of data for multifunctional materials. Currently, researchers utilize transfer learning to address this problem; for example, Gopakumar et al. [76] accumulate knowledge related to high permittivity and bandgap, creating opportunities for developing high dielectric property materials. However, machine learning is still limited by the lag in data, creating barriers to testing or training machine learning models.

2.2. Dataset collection from literature

Search "Web of Science" for 36,715 pieces of literature correlated with "polymer & energy storage." The research is highly popular and the sum of papers will continue to grow. It is clear that this approach is not limited by existing databases and enables researchers to design polymer matrix dielectrics with specific functions, which has been adopted by many researchers [77–79]. Our research group [80] collected 1,254 data sets from 233 pieces of open literature, including fillers, structures, and energy storage properties. Among them, there are more than 50 types of fillers, and the dimensions of fillers range from 0 to 2 dimensions. Chen et al. [66] amassed 1210 experimental permittivity figures for 738

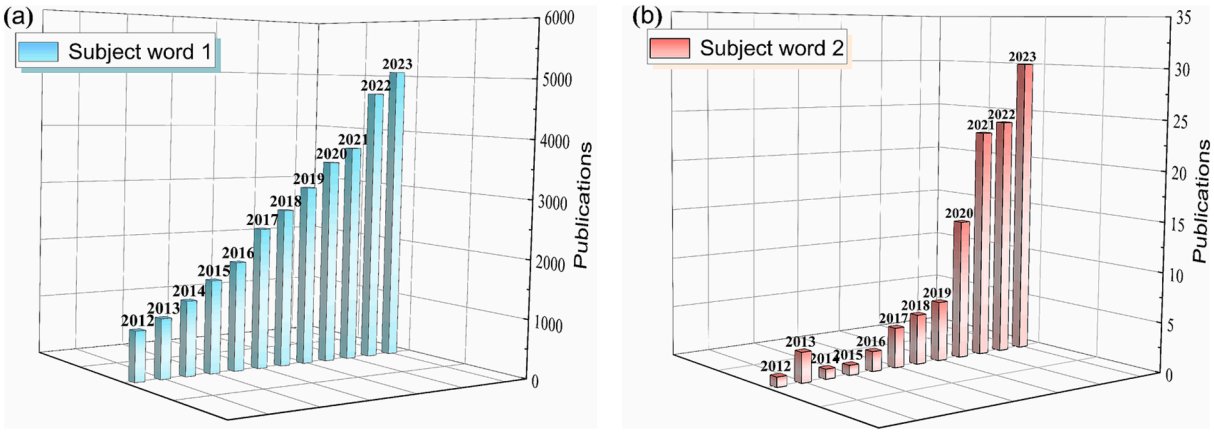


Fig. 6. Number of annual publications in Web of Science related to (a) subject word 1 "polymers, energy storage", (b) subject word 2 "polymers, energy storage, machine learning" from 2012 to 2023.

diverse polymers sourced from 12 literature for training models, where the frequency range includes 60, 10^2, 10^3, 10^4, 10^5, 10^6, 10^7, 10^9, and 10^{15} Hz. Zhu et al. [77] collected data from 52 existing literature on 16 common polymers and 14 nanofillers in uniformly filled composite materials to construct a dataset. The sample numbers of E_b, ε_r, and U_e were 279, 334, and 172, respectively. However, this method takes a lot of time to collect, and the collected data points are not comprehensive compared with the available databases, and there are problems such as data quality and deviation. Some researchers, to improve the collection efficiency, develop filtering software that can automatically extract the relevant information from the papers and finally materialize the dataset from the literature, and this method may be an effective means to break the limitation of collecting data based on the literature [81]. However, it is still challenging to integrate the data points reasonably collected from the text when applying it, and the extracted information can only represent the results of selected filters, effective information needs to be refined to mitigate the risk of model overfitting and enhance its predictive versatility.

2.3. Dataset collection from high throughput computing

In some cases, data points in existing databases or literature cannot meet the needs of researchers; thus, researchers construct datasets through computer simulation methods [82,83]. Our research group [84] constructed a machine learning dataset for constructing the E_b relationship between microstructure and polymer matrix composite by performing 504 sets of high-throughput stochastic breakdown simulations with filler permittivity ε_r, filler size d, and filler content v as the variables. Aiming to explore different forms of nanofillers on the breakdown strength, Shen et al. [67] calculated 6615 combinations to construct a dataset as a training set for machine learning using high-throughput phase field simulations. Five morphologies with a volume fraction of 1 vol% are considered, which are 0D nanoparticles, 1D perpendicular nanofibers, 1D parallel nanofibers, 2D perpendicular nanosheets, and 2D parallel nanosheets. The calculations include morphology S_i, dielectric constant ε_j, conductivity σ_k, and volume fraction V_l.; 21 dielectric constants varying from 3 to 7000, and 21 conductivities varying from 1×10^{-16} S m^{-1} to 1×10^{-6} S m^{-1}. By using three modules for phase-field simulation, the corresponding effective property data $P\left(\varepsilon_{ijkl}^{eff},\ E_{ijkl}^{b},\ \sigma_{ijkl}^{eff}\right)$ of the nanocomposites are output, for a total of 6615 combinations. Aiming to investigate how various types of nanofillers influence the dielectric breakdown strength, Shen et al. [85] calculated the breakdown values of 343 different nanocomposites through high-throughput phase field simulation, thereby establishing a database, and chose three features, e.g., dielectric constant ε, conductivity σ, and Young's modulus Y as the fingerprints to distinguish between different fillers, and combined the twelve prototype functions (x, x^{-1}, $x^{1/2}$, x^2, x^{-2}, x^3, x^{-3}, $\ln(x)$, $1/\ln(x)$, $\exp(x)$, $\exp(x)$) with one (ε, σ, Y) or two (ε & σ, ε & Y, σ & Y), or three (ε & σ & Y) fingerprints to form new compound descriptors for facilitating the subsequent algorithmic analysis. In the process of parameter analysis, the calculation results of this method may be affected by the selected model, and there may be deviations in the results. When further improving its accuracy, more computing resources are needed as support, so compared to other methods, it will consume more computing equipment and time.

2.4. Dataset collection from experimental data

To meet the demand for training in modeling multifunctional polymer-based dielectric, some researchers themselves specialize in designing experimental procedures and measuring experimental data to fill in the gaps of data collected by other means [86,87]. However, the performance is influenced by a constellation of factors, e.g., the type of matrix material, filler distribution orientation, preparation process, and environmental factors, and the time spent on synthesizing and preparing them far exceeds that of the three methods mentioned above. For example, Mukherjee et al. [86] synthesized PVDF/$Ba_{0.7}Sr_{0.3}Zr_{0.02}Ti_{0.98}O_3$(f-BSZT)/$f_1$-MWCNTs nanocomposites through the experiment. The synthesis process parameters were optimized and compared in view of macro and micro characterization. Finally, the energy storage density of this composite reached 14 J/cm^3 (an increase of 204 % compared to 4.6 J/cm^3 of the original PVDF). However, this process demands substantial time and expense due to the complex preparation of raw materials, the doping of 5 different fillers with different volume fractions, and the combination of various characterization methods. Therefore, from the point of view of engineering application, this can only be suitable for the laboratory research and development stage, and cannot be applied in the large-scale market.

2.5. Comparison of disadvantages of different collection methods

For the convenience of understanding the four collection methods mentioned above, a visual comparison and organization are presented in Table 1.

3. Machine learning algorithm optimization

Machine learning algorithms are the core of the entire workflow; they are the key technology to achieve from data to intelligent decision-making and prediction. Therefore, the choice of algorithm may have an impact on the accuracy and efficiency of the model. The models of ML include four categories: supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning [88–90]. At present, supervised learning is widely used, which is usually used for classification and regression. The function of regression is to estimate the value of the output, i.e., minimize the error of the fitting curve on the training data during the training process [91,92]. Aiming to calculate the different identifiers of polymer-based dielectrics, researchers can choose one or more appropriate ML algorithms. Therefore, founded on the most typically used algorithms in the study, four representative algorithms are selected, e.g., least squares regression (LSR), back-propagation neural network (BPNN), Gaussian process regression (GPR), and random forest (RF).

3.1. Least squares regression (LSR) algorithm

LSR is a regression algorithm used in machine learning and statistics

Table 1
Comparison of different dataset collection methods.

Method	Advantages	Disadvantages
Dataset collection from public databases and platforms	1) Data points are relatively plentiful. 2) The acquisition method is convenient.	1) Data has lagged. 2) Incomplete data on multifunctional polymers.
Dataset collection from literature	1) The number of relevant literature is increasing, and breaking the existing database limitations. 2) Data with high reliability.	1) The efficiency of obtaining information is low and time-consuming. 2) Existing issues such as data bias in the collected data.
Dataset collection from high-throughput computing	1) Not constrained by existing databases, more suitable for training data points of multifunctional polymers. 2) Able to predict non-existent data.	1) The calculation process consumes a large amount of computing equipment and time. 2) The calculation accuracy is limited by computing resources and does not have universality.
Dataset collection from experimental data	1) Having more targeted and multifunctional polymer data, the scope is more comprehensive, breaking through the limitations of existing data.	1) Extremely time-consuming and costly, and greatly influenced by external conditions.

to solve linear regression problems [93,94]. For example, Shen et al. [85] first performed high-throughput phase field simulations that included the physical parameters of the polymer-based nanocomposites. The input variables include the dielectric constant ε, which corresponds to the dielectric properties of the filler and matrix, the conductivity σ of the filler and matrix, and the Young's modulus *Y*, which is the mechanical strength variable of the filler and matrix. Conducted regression analysis on 343 breakdown strength databases using LSR with coefficient of determination (R^2) as the screening criterion. The database is exhibited in Fig. 8(a). In order to more accurately describe the relationship between the input variables and the failure strength, 12 nonlinear prototype functions were constructed, including: the linear term x , the reciprocal term x^{-1}; the square root term $x^{1/2}$, $x^{-1/2}$; the quadratic term x^2, x^{-2}; the cubic term x^3, x^{-3}; the logarithmic term ln(x), 1/ln(x) and the exponential term exp(x), exp(−x). To further account for the synergies between the variables, the following interaction terms were constructed: ε and σ, ε and Y, σ and Y, and higher order interaction combinations of the three. To obtain a highly accurate and simple expression, the LSR algorithm performs three rounds of screening. In the first round of screening, the interaction terms between variables are ignored and only the univariate terms are retained, and the goodness-of-fit R^2 of the regression model is calculated as 0.8899. In the second round of screening, the interaction terms between variables are introduced and the regression model is retrained to improve the accuracy of the prediction, and the R^2 is 0.9099. In the third round of screening, the variables and their interaction terms that have the greatest influence on the breaking strength are screened out to ensure the simplicity and practicality of the formula. term. After three rounds of regression analysis using three fingerprints, four interactions, and twelve prototype functions, machine learning was used to generate the expression for the breakdown strength of (P(VDF-HFP))-based nanocomposites. Based on practical problems, the analytical formula will only consider the interplay between conductivity σ and Young's modulus Y is exhibited in Eq. (7). The coefficient of determination R^2 is 0.9093.

$$\frac{E_b^{composite}}{E_b^{matrix}} = 0.9058 - 0.01175\ln\frac{\sigma_{filler}}{\sigma_{matrix}} - 0.06767\ln\frac{\sigma_{filler}}{\sigma_{matrix}} - 0.01640\left(\frac{Y_{filler}}{Y_{matrix}}\right)^{-1/2} + 0.001\left(\frac{Y_{filler}}{Y_{matrix}}\right)^{-1/2}\ln\frac{\sigma_{filler}}{\sigma_{matrix}} \quad (7)$$

For the growth of *E*composite b, ε_{filler}, and σ_{filler} have a negative effect, while Y_{filler} has a positive effect. Ceramic materials have a much higher Young's modulus than polymer matrices, therefore, it is crucial to find nanofillers with low conductivity to improve the breakdown strength. To test the reliability of phase field simulation and analytical expressions, compare them with available experimental measurement data. As exhibited in Fig. 8(b), there are 343 training dates, of which 7 data points have been marked with 7 colors outside the training dataset, namely Al_2O_3, TiO_2, SiO_2, $BaTiO_3$, $SrTiO_3$, MgO, and $Ba_yCa_{1-y}(Zr_xTi_{1-x})O_3$(BCZT). These 7 colored data points are distributed around the solid line, indicating that the phase field simulation calculation of the breakdown strength is consistent with the prediction of machine learning, thus verifying the reliability of the analytical formula. To verify the accuracy, P(VDF-HFP) powder was thoroughly dissolved in a solvent mixture of N,N-dimethylformamide and acetone. The Al_2O_3 nanoparticles were dispersed in the P(VDF-HFP) solution by ultrasonication at 5 % volume fraction. Electrostatic spinning and hot pressing processes were carried out to obtain the final Al_2O_3/P(VDF-HFP) nanocomposites with a thickness of about 15 μm. The breakdown strengths of the prepared pure P(VDF-HFP) polymer and Al_2O_3/P(VDF-HFP) nanocomposites were measured, with breakdown strengths of 561.2 kV/mm for the pure P(VDF-HFP) polymer and 647.6 kV/mm for the Al_2O_3/P(VDF-HFP) nanocomposite. A strategy for the optimal selection of nanofillers to improve the breakdown performance of polymer-based dielectric materials is proposed, providing important guidance for theoretical and experimental researchers of high energy density dielectric materials and devices.

Moreover, our research group [84] constructed a machine learning database by generating 504 datasets using a high-throughput stochastic breakdown simulation method with dielectric constant (ε_r), packing size (d) and packing content (v) as variables and breakdown strength (E_b) as the target value. The dataset is shown in Fig. 9(a-h). Normalisation of ε_r and E_b is performed to ensure that the results are applicable to different ranges of polymer materials. The three input variables are defined as x_1 instead of dielectric constant (ε_r), x_2 instead of filler size (d) and x_3 instead of filler content (v). To better describe the non-linear relationships, 12 prototype functions and 4 interaction terms between the variables were considered. Conducted regression analysis using LSR based on a dataset consisting of E_b values of 504 different polymer-based dielectrics. The coefficient of determination (adjusted-R^2) of the LSR was used for culling descriptors. In the first round of screening, the optimal function was filtered by regression analysis using a function for each variable (x_1, x_2, x_3) separately and sorted by adjusted-R^2. The optimal

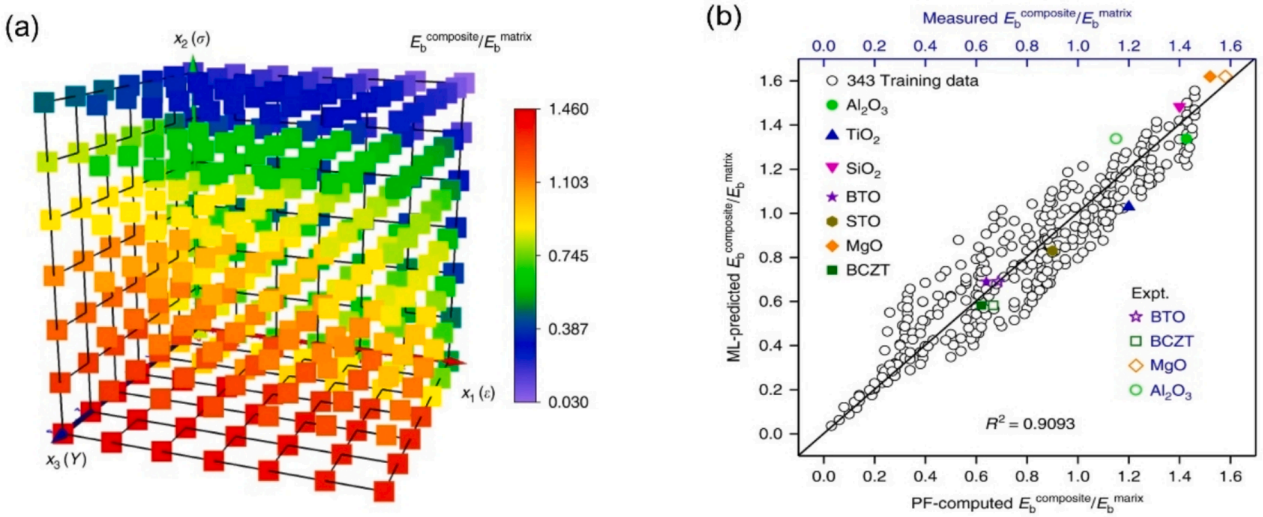


Fig. 8. (a) A high-throughput phase field simulation breakdown strength dataset of 5 vol% nanofiller filled P(VDF-HFP) composites with different dielectric constants, conductivity, and Young's modulus. (b) Comparison of breakdown strength between PF simulation and ML prediction. Reproduced with permission. [85] Copyright 2019, Springer Nature Limited.

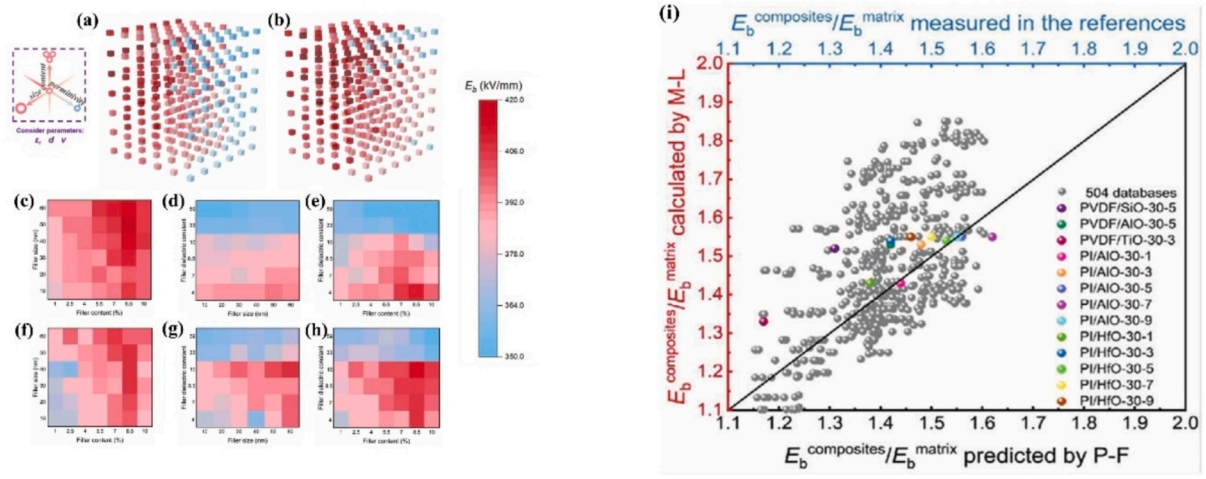


Fig. 9. High throughput random breakdown simulation. The breakdown strength of (a) PI matrix (b) PVDF matrix with three variables, i.e. the filler sizes (d) (from 10, 20, 30, 40, 50, 60 nm), the filler contents (v) (from 1, 2.5, 4, 5.5, 7, 8.5, 10 vol%), and the filler ε_r (from 4, 7, 8.5, 10, 33, 59); two variables selected from three variables (d, v, ε_r) as the E_b 2D mapping of functions PI (c)-(e) and PVDF (f)-(h). (i) Standardized comparison between random breakdown simulation and machine learning prediction. Reproduced with permission. [84] Copyright 2022, John Wiley and Sons.

function for x_1 was found to be $x^{-1/2}$ with an R^2 of 0.743; for x_2, the optimal function was x^2 with an R^2 of 0.007 and for x_3, the optimal function was e^{-x} with an R^2 of 0.009. Combining the first three functions of x_1 with the optimal functions of x_2 and x_3 for both bivariate and trivariate regression analyses showed that the combination of x_1-ln(x) and x_2- x^2 had a higher R^2 of 0.34, and the three-variable combination of x_1-ln(x), x_2- x^2 and x_3-e^{-x} had the best-fitting R^2 of 0.042. For the third round of screening, an interaction term is introduced between the variables (e.g., x_1 & x_2, x_1 & x_3). Through three rounds of regression analysis, the optimal function describing each variable is gradually determined, and the best adjusted $R^2 = 0.778$ is obtained. Owing to the complex effects of the dielectric constant ε_r, filler size d, and filler content v on the breakdown strength E_b, this is an acceptable result, as expressed in Eq. (8).

$$Y = 1.455 - 0.089(x_1)^{-1/2} + 0.00012(x_2)^2 - 0.156e^{-(x_3)} + 0.000006\ln(x_1)(x_2)^2 - 0.061\ln(x_1)e^{-(x_3)} \quad (8)$$

According to the results of the ML model and the random breakdown simulation, as exhibited in Fig. 9(i). The data points distributed near the black line indicate a high degree of consistency in the results produced by the two methods. For data points above the black line, validated with $R^2 = 0.778$, it is close to 1. In addition, colored symbols represent the use of additional literature [95,96] to validate the reliability of machine learning. To verify the accuracy, polyetherimide (PEI) was chosen as the polymer matrix and Al_2O_3 (alumina) nanoparticles with different volume fractions were used. The experimental results showed that the breakdown strength values of pure PEI and composites with 1 vol%, 3 vol% and 5 vol% Al_2O_3 were 363, 474, 509 and 529 kV/mm, respectively. The simulated breakdown strength Eb of the PEI/Al_2O_3 The simulated breakdown strength E_b of the composites were 464, 492 and 517 kV/mm (volume fraction of Al_2O_3 was 1 vol%, 3 vol% and 5 vol%, respectively). The deviation between the simulated and experimental results is less than 3.5 %, which is within acceptable limits. The results of both experimental tests and stochastic fracture simulations show a consistent trend of increasing fracture strength E_b of PEI/Al_2O_3 composites with increasing Al_2O_3 content.

The principle of this method is simple, the solving process is efficient, and it is suitable for various linear regression problems, including univariate and multivariate. When there are too many polymer features, it is necessary to avoid overfitting. Meanwhile, it is sensitive to outliers, which may affect the accuracy of the model.

3.2. Backpropagation neural network (BPNN) algorithm

BPNN is a multi-layer feedforward artificial neural network that is trained using supervised learning and optimized for weights and biases through a backpropagation algorithm [97]. For example, Shen et al. [67] first constructed a data set through high-throughput phase field simulation as a training model for machine learning. The input set I(S_i, ε_j, σ_k, V_l) consists of four main characteristics of nanocomposites: morphology S_i, dielectric constant ε_j, conductivity σ_k, and volume fraction V_l. Through the three-module phase-field simulation, we are able to output the corresponding data set of nanocomposites, including effective dielectric constant ε_{ijkl}^{eff}, breakdown strength E_{ijkl}^{b}, and effective conductivity σ_{ijkl}^{eff}. Here, we consider five morphologies, 21 dielectric constants (from 3 to 7000), 21 conductivities (from 1×10^{-16} S·m^{-1} to 1×10^{-6} S·m^{-1}), and a fixed volume fraction of 1 %, resulting in a total of 6615 combinations. Then, taking into account the contribution of the conductive loss to the discharge energy density at high electric fields and based on a linear approximation, a scoring function was proposed to evaluate the energy storage capacity of different nanocomposites, as expressed in Eq. (9).

$$f_s = 1/2\left(E_{ijkl}^{b}\right)^2\left(\varepsilon_0\varepsilon_{ijkl}^{eff} - \sigma_{ijkl}^{eff}\Delta t\right) \quad (9)$$

where Δt is the period of the applied electric field. Using the min–max normalisation method, we normalised the values of the scoring dataset to values between 0 and 1, as shown in Fig. 10a. The closer the scoring value is to 1, the higher the energy storage capacity of the material. Based on the established scoring dataset as the training set, the backpropagation neural network (BPNN) algorithm was used to predict the material properties by ML. Using the BPNN backpropagation neural network as the machine learning model, the dataset was divided into a training set (90 %) and a test set (10 %). The non-linear relationship between the input layers of the four input features and the energy storage score fs output was automatically learned by the neural network. Fig. 10(b) shows the comparison of ML results f_s^{ML} and PF scoring results f_s^{PF} for all datasets (including 90 % of the training set and 10 % of the testing set), marked with hollow symbols. All the hollow symbols scattered around the black line represent $f_s^{PF} = f_s^{ML}$, indicating a high degree of approximation between ML prediction and PF simulation, verifying its high reliability. The linear correlation between the predicted and simulated values of the model was assessed using the Pearson correlation coefficient (PCC). The mean square error (MSE) was used to

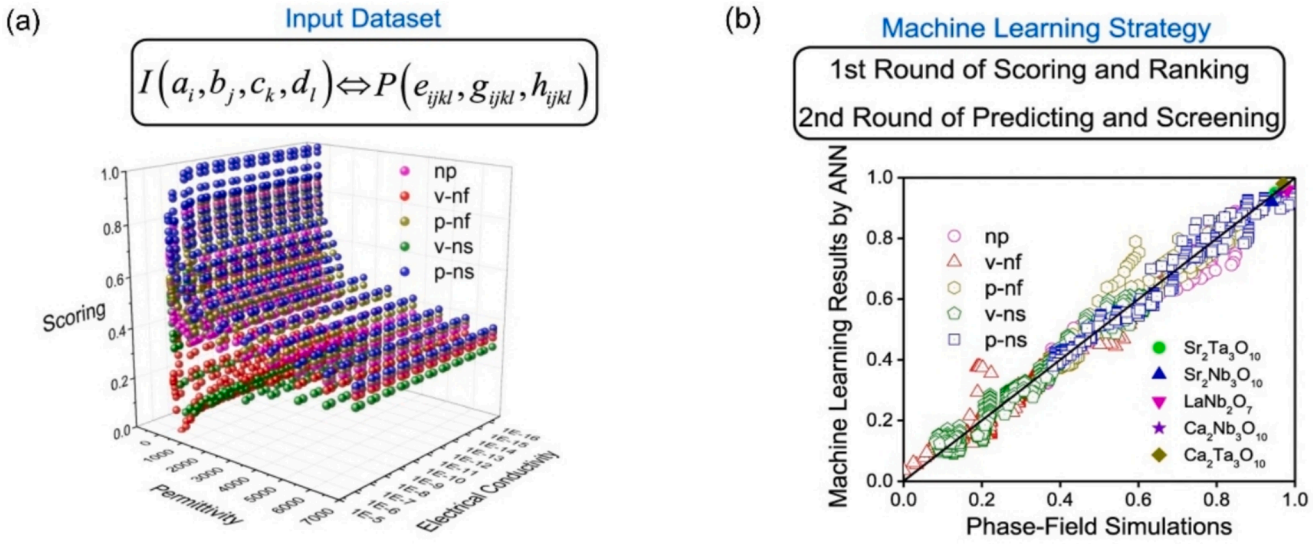


Fig. 10. (a) Input dataset for scoring different polymer nanocomposites. (b) Comparison between machine learning results and phase field score simulations for all datasets, with colored solid symbols representing evaluations of five types of two-dimensional perovskite nanofillers. Reproduced with permission. [67] Copyright 2021, Springer Nature Limited.

measure the difference between the predicted results and the true values. As expressed in Eq. (10) and Eq. (11).

$$PCC = \frac{N\sum(x_i y_i) - \sum x_i \sum y_i}{\sqrt{N\sum x^2 - (\sum x_i)^2}\sqrt{N\sum y^2 - (\sum y_i)^2}} \tag{10}$$

$$MSE = \frac{1}{m}\sum_{i=1}^{m}\left(y_i - y_i^{'}\right)^2 \tag{11}$$

The PCC and MSE are 0.97 and 9.01×10^{-4}, respectively, suggesting that the ML model has high precision. By culling existing two-dimensional materials, it was found that some two-dimensional nanosheets composed of TiO_6, NbO_6, or TaO_6 octahedral perovskite structure blocks, such as $Sr_2Ta_3O_{10}$, $Ca_2Nb_3O_{10}$, $LaNb_2O_7$, $Sr_2Nb_3O_{10}$, and $Ca_2Ta_3O_{10}$. The solid symbols exhibited in Fig. 10(b) have received very high rating scores, demonstrating high potential for development with high energy capacity. The selection of P(VDF-HFP)/$Ca_2Nb_3O_{10}$ polymer nanocomposites as the target material in the above predicted 2D chalcogenide nanosheets has verified the predicted conjecture. The introduction of $Ca_2Nb_3O_{10}$ nanosheets at a volume fraction of 0.1 % into the P(VDF-HFP) matrix resulted in a 31 % increase in the fracture strength from 649 MV/m to 853 MV/m for the original P(VDF-HFP). Finally, the data obtained by combining machine learning prediction and experimental testing is included in the database, and the database data is optimized for iterative updates. Phase field modelling and machine learning strategies can also be extended to study the energy density properties of other dielectric materials, including mixed polymers, block copolymers, semi-crystalline polymers and multiphase ceramics.

This method has high prediction accuracy for breakdown strength and is sensitive to initial weights. Therefore, careful initialization is necessary to avoid overfitting when dealing with insufficient data.

3.3. Gaussian process regression (GPR) algorithm

GPR is a nonparametric machine learning technique based on Gaussian process theory, which has the advantage of exhibiting excellent prediction accuracy on small datasets and providing uncertain information about the model output [98,99]. For example, Zhu et al. [77] collected experimental data from existing literature, including values of dielectric breakdown strength E_b, dielectric constant ε_r and energy storage density U_e. The data included 16 common polymers and 14 nanofillers. All data were subjected to min–max normalisation to scale the data between 0 and 1 in order to eliminate the effect of data dimensional inconsistency on model training and to improve model convergence speed and stability. Physical parameters (e.g. dielectric constant, band gap, etc.), geometrical structure (e.g. shape, distribution, orientation) and other parameters of the nanofillers were used for fingerprinting. Key features that dominate the dielectric properties are identified through feature extraction and analysis. Utilized the GPR technique to discover the link between the microstructural features and dielectric properties (E_b, ε_r, and U_e) of composite materials. They used five-fold cross-validation in dataset segmentation to prevent overfitting and evaluated the predictive accuracy of the ML model through RMSE, MAE, and R^2. Fig. 11(a)-(c) shows the parity check graph between the ML predicted values and experimental values of E_b, ε_r, and U_e, respectively. They indicate that using ML models to classify the relationship is relatively accurate, and the relative error of most test data is less than 20 %. As exhibited in Fig. 11(d), the RMSEs of E_b and ε_r are 0.126 and 0.178, respectively, and R^2 is 0.802 and 0.877, respectively. However, some data points result in significant prediction errors for U_e, with RMSEs and R^2 of 3.09 and 0.634, respectively, which may be attributed to the limited number of U samples. To further validate the accuracy of the ML model, as exhibited in Fig. 11(e), using three types of polymer-based composite materials to validate, e.g., P(VDF-TrFE-CFE)/BNNS (2D filler), P(VDF-HFP)/TiO_2(1D filler), and P(VDF-TrFE-CFE)/$BaTiO_3$(0D filler). It can be seen that the predicted E_b of the three different composite materials with changes in volume fraction is consistent with the actual experimental trend。The predicted E_b values of P(VDF-TrFE-CFE)/BNNS first increase and then decrease with increasing volume fraction, and the predicted E_b values of P(VDF-HFP)/TiO_2 and P(VDF-TrFE-CFE)/$BaTiO_3$ decrease with increasing volume fraction, which is in agreement with the experimental trend. This shows that the ML model can well describe the relationship between fingerprint and dielectric properties and can be used to guide the rational design of polymer composites with desirable properties. Results suggesting that the ML model can well describe the relationship between microstructure and dielectric properties.

This method requires professional knowledge and extensive experimentation to select appropriate kernel functions, reducing the impact on model performance. Due to the high computational complexity when processing large-scale datasets, time preparation is required when using it.

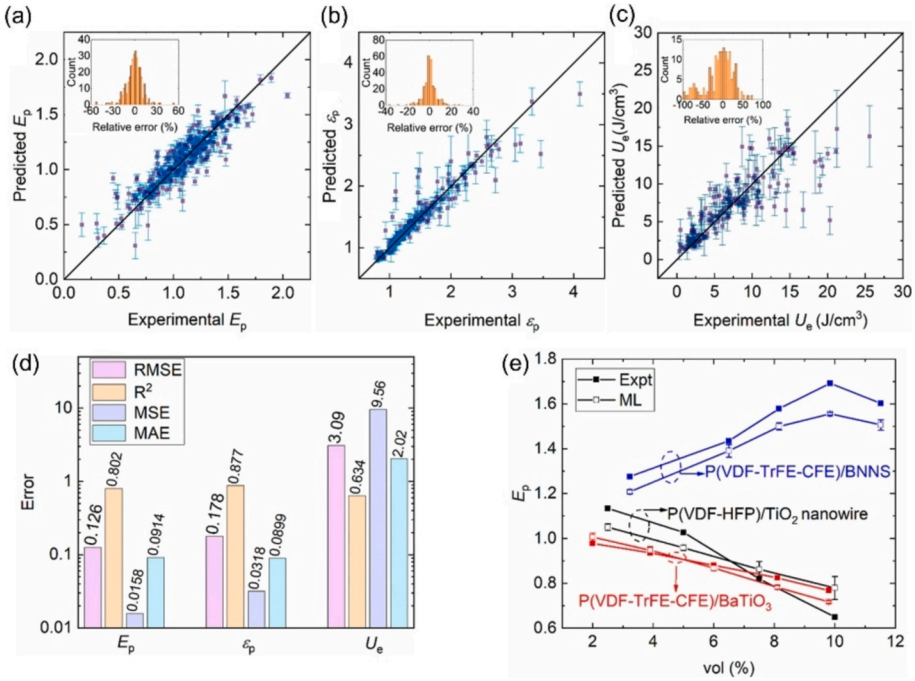


Fig.11. The parity graph between ML prediction and experimental values for (a) Breakdown strength. (b) Dielectric constant. (c) Energy density. (The yellow cylindrical distribution chart represents the distribution in the dataset.) (d) The prediction error of the ML model. (e) The result comparison of experimental and predictive breakdown strength with volume fraction variation. Comparison expression: $E_p = E_{bcomposite}/E_{bmatrix}$, $\varepsilon_p = \varepsilon_{composite}/\varepsilon_{matrix}$. Reproduced with permission. [77] Copyright 2021, American Chemical Society. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.4. Random forest (RF) algorithm

RF is an ensemble learning method that constructs multiple decision trees for classification or regression problems and combines their predicted results to improve the accuracy of the model [100,101]. Our research group found that the degree of filler dispersion in the matrix is also an important factor affecting the maximum energy storage density. Therefore, to explore the impact of filler dispersion on prediction accuracy, the research group [80] selected 869 sets of data from the 1254 data sets, including composite material microscopic image information (SEM images), a dataset of filler content, and energy performance, as exhibited in Fig. 12(a). Firstly, the SEM images are processed by setting all images to grayscale format using Photoshop and then converting the SEM image information into a 200 × 200 matrix format using Anaconda software. Considering the simplification of operations, the nonnegative matrix factorization (NMF) method is used to reduce the dimensionality of the data and convert it to a size of 200 × 1 [102,103]. Combining image information, an ML prediction model is established using RF, SVM, and NN algorithms. The results of three algorithms predicting the accuracy of two test sets are exhibited in Fig. 12(b). Red represents the absence of SEM image data in the dataset, while blue represents the presence of SEM image data. By comparing and analyzing the prediction accuracy of each algorithm under two test sets, the group found that adding SEM image data to the dataset can improve the prediction accuracy. The RF algorithm with the best accuracy increased from 84.1 % to 91.9 %. The distribution of fillers added to the dataset will have an

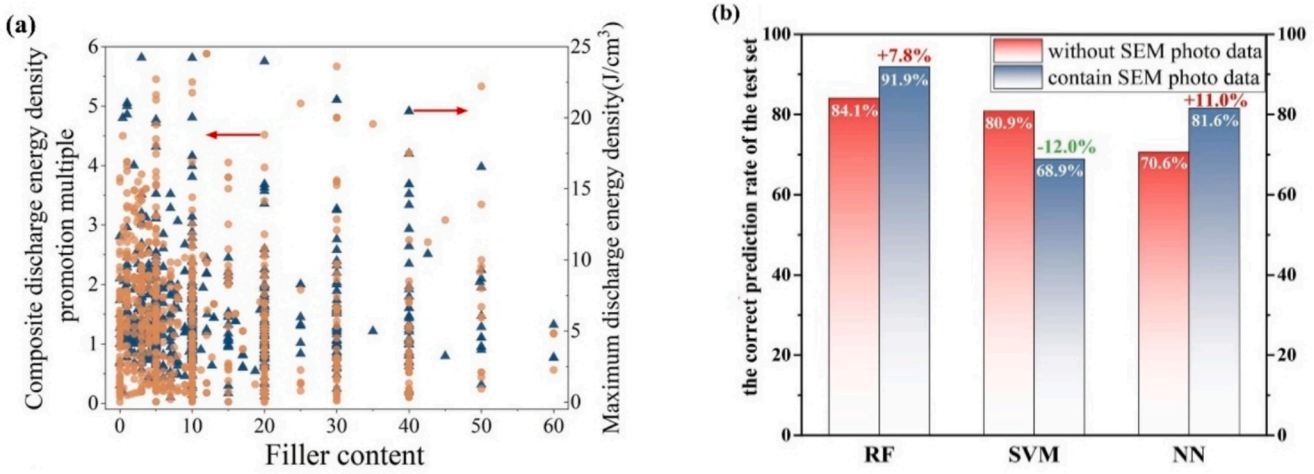


Fig. 12. (a) The relationship between filler content and composite material energy performance. (b) The prediction accuracy of three algorithms in the test set with and without SEM image data. Reproduced with permission. [80] Copyright 2021, John Wiley and Sons.

impact on the prediction model, indirectly inferring that the dispersion degree of fillers in the matrix will affect the energy performance of composite materials. Among the three algorithms, the RF algorithm has a better prediction accuracy, while the phenomenon of the SVM algorithm's prediction accuracy decreasing after adding SEM image data may be due to the principle of the SVM algorithm being to find the optimal boundary between data points, thus too many input descriptors will indeed reduce the prediction accuracy. To verify the reliability of the predictive model, monolayer and multilayer structured composites were prepared by high-speed directional spinning using boron nitride nanosheets (BNNS) and poly (vinylidene fluoride–trifluoroethylene–-chlorotrifluoroethylene). The volume fractions of fillers in the monolayer composite films were 1 %, 2 %, 3 %, 4 % and 5 %, respectively. The multilayer gradient structure contained 10 layers, which were classified into four types according to the different volume fractions of the outer-middle-outer layers and named 0–8-0, 1–7-1, 2–6 −2 and 3–5-3, and the gradient of each layer corresponds to 2, 1.5, 1 and 0.5. The descriptors in the experimental design (filler volume fraction, structure type, etc.) are used as inputs to predict the maximum energy storage density of the composites. If the maximum energy storage density of the composite is at least 1.2 times that of the matrix, it is recorded as 'R' (successful prediction). The experimental results are in good agreement with the model predictions and the model is able to accurately predict the range of maximum energy storage densities of composites. In particular, the Random Forest (RF) model performs best, accurately predicting the performance improvement for different volume fractions and structures.

This method is relatively insensitive to parameter selection but still requires adjusting to some parameters, such as the number of trees. When there are many numbers, it may occupy a large amount of memory space, and as a whole model, it needs to improve the interpretability of the decision-making process.

3.5. Comparison of advantages and disadvantages of different algorithms

In order to compare the advantages and disadvantages of each algorithm more intuitively, they will be listed through a table with the specific information shown in Table 2.

4. Preface to machine learning dynamics

4.1. Breaking through the constraints of small datasets

Machine learning methods typically rely on large-scale datasets to achieve good performance. In fields such as image recognition, the ease of accessing millions of data points makes this issue less apparent. However, in fields such as chemistry and materials science, data acquisition often faces challenges, typically with only a few hundred or thousands of high-quality data points, or even fewer. Therefore, researchers urgently need to make experimental results and data publicly available in a computer-readable format to improve data sharing and usability.

Table 2
Comparison of different algorithms.

Algorithm	Advantages	Disadvantages
LSR	1) The principle of the method is simple. 2) Efficient solution process with wide applicability	1) prone to overfitting. 2) Sensitivity to outliers
BPNN	1) Highly predictable. 2) Adaptable and effectively predictable within data conditions.	1) High initialisation requirements, the setting of initial weights can significantly affect model performance.
GPR	1) Optional kernel function to effectively improve model performance. 2) Highly flexible.	1) Dependence on expertise and high computational complexity. 2) In big data scenarios, sufficient time needs to be set aside for model training.
RF	1) The model is less dependent on parameters and has better overall robustness. 2) The performance can be optimised by adjusting the parameters with some flexibility.	1) Large memory footprint, the model consumes a lot of memory space. 2) The interpretability of the decision-making process needs to be improved.

To address the challenges of small datasets, meta learning provides an effective solution by extracting and transferring knowledge between different tasks to improve model learning efficiency. For example, technologies such as neural Turing machines and imitation learning are gradually driving the development in this direction. In addition, the latest research shows that methods based on Bayesian frameworks have reached a level comparable to humans in a single learning task [104]. This progress is particularly important for molecular and materials science, as data acquisition in these fields is costly, time-consuming, and resource limited.

4.2. Innovation and optimization of algorithms

Deep learning (DL) is a subfield of machine learning that refers to enabling computers to learn from data and perform tasks by modelling the structure and functionality of neural networks (especially multilayer neural networks). Deep learning is characterised by the use of multi-layer (deep) structures to process complex input data, automatically extracting features and thus eliminating the need for manual feature engineering. Key techniques include Convolutional Neural Networks (CNNs) [105], which are mainly used for image processing and are good at extracting local features from images; Recurrent Neural Networks (RNNs), which are used to process sequential data, such as time series or text, and are able to remember previous states; and Generative Adversarial Networks (GANs) [106], which are used to generate new data, such as images or text, and contain a generator and a discriminator that improve the quality of the generated data through adversarial training.

Compared to traditional algorithms such as Random Forest and Gaussian Process Regression, where Random Forest is insensitive to feature selection and relatively computationally efficient. Gaussian Process Regression is a nonparametric generalised approach suitable for small sample learning that can model nonlinear relationships through kernel functions. Deep learning relies on large amounts of labelled data and computational resources, and is prone to overfitting when the amount of data is insufficient. It is widely used for unstructured data (images, files, audio), such as image recognition, natural language processing, and recommender systems, etc. However, it is a black box model and is difficult to apply to all types of data, such as images, text and audio. However, being a black box model, it is difficult to explain the specific decision process. Therefore, if the data volume is small and interpretability is important, Random Forest or Gaussian Process Regression can be chosen. If the data volume is large, the problem is complex, and the performance requirements are high, deep learning is usually the optimal choice, especially for unstructured data analysis [107].

4.3. Optimisation models

Hyperparameter tuning occupies a very important position in the optimisation process of machine learning models, and it is a key step in the optimisation process. Among them, hyperparameter selection directly affects the model performance and training effect. Hyperparameter selection methods and strategies, including:Manual adjustment based on experience [108], which manually sets hyperparameters based on experience or recommended values in the literature, and then gradually adjusts them through experiments; Grid search [109], which specifies a set of possible values of hyperparameters and selects the optimal hyperparameters by exhaustive trials of all combinations; Random search (Random search) [110], which randomly samples a certain number of combinations from the hyperparameter space for

evaluation; Bayesian optimization) which intelligently selects the next combination of hyperparameters based on the probability model and the collection function (e.g., Gaussian process), and progressively approximates the optimal solution and Auto-adjustment tool [111], which uses existing autotuning tools such as AutoML, Optuna, etc. for hyperparameter selection.

In machine learning, hyperparameter tuning has a profound impact on model performance and practical applications, which includes the following aspects: model performance improvement, by optimising hyperparameters, the accuracy, generalisation ability and overall performance of the model can be effectively improved, and the training error and testing error can be reduced. Different combinations of hyperparameters directly affect the training process of the model, such as the learning rate, regularisation coefficient and tree depth. Appropriate tuning of the hyperparameters helps to balance the complexity of the model and prevent overfitting (overlearning of the training data) or underfitting (underlearning of the training data). Different algorithms (e.g. decision trees, support vector machines, neural networks, etc.) have different sensitivities to hyperparameters, and tuning hyperparameters can significantly improve the performance of a particular algorithm. Model fitness on different datasets also depends on the choice of hyperparameters. In real-world scenarios, optimised hyperparameters can improve the use of models, e.g. by reducing inference time and improving real-time response time [112]. Therefore, accurate hyperparameter tuning is particularly important for resource-constrained devices (e.g. embedded systems).

5. Summary and outlook

In summary, the assistance of ML has promoted the research of energy storage polymer-based dielectric. This review summarized the relevant published works and development trends of ML in materials research in recent years. Compared with traditional experiments, it has significant advantages in processing large amounts of data, optimizing filler selection, predicting performance, and discovering new materials. In addition, by analyzing the excellent achievements of researchers in the field of polymer-based dielectrics research, a deep summary has been made of data collection and algorithm optimization. The ultimate proof is that ML helps reduce the research and development costs of polymer energy storage dielectrics, shortening the research and development cycle, and improving the performance and manufacturing efficiency of materials. However, some potential issues still pose challenges to ML. Therefore, the following outlook is proposed.

(1) Construct a High-quality Database

The breadth and reliability of data have always been one of the biggest obstacles in machine learning. Faced with the development of multifunctional polymer-based dielectrics, the existing databases have limited effective information, resulting in a significant decrease in the predictive ability and accuracy of the models. In the future, establishing a comprehensive database based on experiments, literature, databases, and computer calculations. Meet the function of users uploading data and automatically updating and retrieving it, thereby improving data quality while reducing uncertainty.

(2) Optimize Descriptors

The selection and construction of descriptors are crucial for building efficient and accurate prediction models. The description of polymer-based dielectrics should cover the size, content, and distribution of fillers, as well as the performance parameters of the matrix and fillers. The existing descriptors may have issues such as information redundancy, high dimensionality, or weak association with target attributes, which may lead to overfitting or insufficient generalization ability of the model. Gain an in-depth perception of the mapping correlation between material data and machine learning models using specialized knowledge in future application fields, refine existing descriptors, and develop more characteristic descriptors.

(3) Increase Model Interpretability

Although machine learning can assist in building relationships between material properties and descriptors, most model algorithms operate in a black box, and the underlying mechanisms behind performance are inexplicable, hindering researchers from gaining a more comprehensive understanding of target characteristics. In the future, enhancing interpretability by improving algorithm maturity and combining it with evaluating instance prediction results will help users better understand and trust the decision-making process.

CRediT authorship contribution statement

Qixin Yuan: Writing – review & editing, Writing – original draft, Supervision, Investigation, Formal analysis, Conceptualization. **Dong Yue:** Writing – review & editing, Resources, Investigation. **Zhe Zhang:** Conceptualization. **Yu Feng:** Formal analysis. **Qingguo Chen:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was funded by the Postdoctoral Fellowship Program of CPSF (No. GZC20230638), China Postdoctoral Science Foundation (No. 2024MD753932), Basic Research Support Program for Outstanding Young Teachers in Heilongjiang Provincial Universities (No. YQJH2023259), Fundamental Research Foundation for Universities of Heilongjiang Province (No. 2023-KYYWF-0114).

Data availability

No data was used for the research described in the article.

References

[1] R.K. Srivastava, N.P. Shetti, K.R. Reddy, E.E. Kwon, M.N. Nadagouda, T. M. Aminabhavi, Biomass utilization and production of biofuels from carbon neutral materials, Environ. Pollut. 276 (2021) 116731.
[2] A.S. Aricò, P. Bruce, B. Scrosati, J.M. Tarascon, W. Van Schalkwijk, Nanostructured materials for advanced energy conversion and storage devices, Nat. Mater. 4 (5) (2005) 366–377.
[3] L. Xiang, Energy network dispatch optimization under emergency of local energy shortage with web tool for automatic large group decision-making, Energy 120 (2017) 740–750.
[4] J. Jurasz, F.A. Canales, A. Kies, M. Guezgouz, A. Beluco, A review on the complementarity of renewable energy sources: concept, metrics, application and future research directions, Sol. Energy 195 (2020) 703–724.
[5] M.M. Rahman, A.O. Oni, E. Gemechu, A. Kumar, Assessment of energy storage technologies: a review, Energy Conv. Manag. 223 (2020) 113295.
[6] L.-T. Yang, X. Kong, F. Li, H. Hao, Z.-X. Cheng, H.-X. Liu, J.-F. Li, S.-J. Zhang, Perovskite lead-free dielectrics for energy storage applications, Prog. Mater. Sci. 102 (2019) 72–108.
[7] Q.-K. Feng, S.-L. Zhong, J.-Y. Pei, Y. Zhao, D.-L. Zhang, D.-F. Liu, Y.-X. Zhang, Z.-M. Dang, Recent progress and future prospects on all-organic polymer dielectrics for energy storage capacitors, Chem. Rev. 122 (3) (2021) 3820–3878.
[8] B.-B. Yang, Y. Zhang, H. Pan, W.-L. Si, Q.-H. Zhang, Z.-H. Shen, Y. Yu, S. Lan, F.-Q. Meng, Y.-Q. Liu, H.-B. Huang, J.-Q. He, L. Gu, S.-J. Zhang, L.-Q. Chen, J. Zhu, C.-W. Nan, Y.-H. Lin, High-entropy enhanced capacitive energy storage, Nat. Mater. 21 (9) (2022) 1074–1080.
[9] H. Luo, F. Wang, R. Guo, D. Zhang, G.-H. He, S. Chen, Q. Wang, Progress on polymer dielectrics for electrostatic capacitors application, Adv. Sci. 9 (29) (2022) 2202438.
[10] H. Pan, S. Lan, S.-Q. Xu, Q.-H. Zhang, H.-B. Yao, Y.-Q. Liu, F.-Q. Meng, E.-J. Guo, L. Gu, D. Yi, X.-R.-S. Wang, H.-B. Huang, J.-L. MacManus-Driscoll, L.-Q. Chen, K.-J. Jin, C.-W. Nan, Y.-H. Lin, Ultrahigh energy storage in superparaelectric relaxor ferroelectrics, Science 374 (6563) (2021) 100–104.
[11] J.-F. Dong, X.-L. Deng, Y.-J. Niu, Z.-Z. Pan, H. Wang, Research progress of polymer based dielectrics for high-temperature capacitor energy storage, Acta Phys. Sin. 69 (21) (2020) 217701.
[12] J.-Y. Pei, L.-J. Yin, S.-L. Zhong, Z.-M. Dang, Suppressing the loss of polymer-based dielectrics for high power energy storage, Adv. Mater. 35 (3) (2023) 2203623.

[13] B.-J. Chu, X. Zhou, K.-L. Ren, B. Neese, M.-R. Lin, Q. Wang, F. Bauer, Q.-M. Zhang, A dielectric polymer with high electric energy density and fast discharge speed, Science 313 (5785) (2006) 334–336.
[14] H.-X. Chen, Z.-B. Pan, Y. Cheng, X.-P. Ding, J.-J. Liu, Q.-G. Chi, M.-H. Yang, J.-H. Yu, Z.-M. Dang, Ultrahigh charge-discharge efficiency and high energy density of a high-temperature stable sandwich- structured polymer, J. Mater. Chem. A 10 (3) (2022) 1579–1587.
[15] Y. Gong, D. Chen, J.-J. Duan, X.-H. Zhang, Y.-H. Ma, C.-W. Zhao, W.-T. Yang, Largely enhanced energy density of BOPP-OBT@ CPP-BOPP sandwich-structured dielectric composites, J. Mater. Chem. C 10 (36) (2022) 13074–13083.
[16] Y. Zhou, C. Yuan, S.-J. Wang, Y.-J. Zhu, S. Cheng, X. Yang, Y. Yang, J. Hu, J.-L. He, Q. Li, Interface-modulated nanocomposites based on polypropylene for high-temperature energy storage, Energy Storage Mater. 28 (2020) 255–263.
[17] J.-L. Li, X.-X. Liu, Y. Feng, D.-Y. Chen, Y.-P. Li, D. Yue, B.-S. Huang, J.-H. Yin, Constructing bidirectional-matched interface between polymer and 2D nanosheets for enhancing energy storage performance of the composites, Energy Storage Mater. 54 (2023) 605–614.
[18] P. Wang, L.-M. Yao, Z.-B. Pan, S.-H. Shi, J.-H. Yu, Y. Zhou, Y. Liu, J.-J. Liu, Q.-G. Chi, J.-W. Zhai, Q. Wang, Ultrahigh energy storage performance of layered polymer nanocomposites over a broad temperature range, Adv. Mater. 33 (42) (2021) 2103338.
[19] G. Liu, Y. Feng, T.-D. Zhang, C.-H. Zhang, Q.-G. Chi, Y.-Q. Zhang, Y. Zhang, Q.-Q. Lei, High-temperature all-organic energy storage dielectric with the performance of self-adjusting electric field distribution, J. Mater. Chem. A 9 (30) (2021) 16384–16394.
[20] Q.-K. Feng, Y.-X. Zhang, D.-F. Liu, Y.-H. Song, L. Huang, Z.-M. Dang, Dielectric and energy storage properties of all-organic sandwich-structured films used for high-temperature film capacitors, Mater. Today Energy 29 (2022) 101132.
[21] Y. Cui, Y. Feng, T.-D. Zhang, C.-H. Zhang, Q.-G. Chi, Y.-Q. Zhang, X. Wang, Q.-G. Chen, Q.-Q. Lei, Excellent energy storage performance of ferroconcrete-like all-organic linear/ferroelectric polymer films utilizing interface engineering, ACS Appl. Mater. Interfaces 12 (50) (2020) 56424–56434.
[22] W.-C. Zhang, F. Guan, M. Jiang, Y.-P. Li, C.-C. Zhu, D. Yue, J.-L. Li, X.-X. Liu, Y. Feng, Enhanced energy storage performance of all-organic sandwich structured dielectrics with FPE and P(VDF-HFP), Compos. Pt. A-Appl. Sci. Manuf. 159 (2022) 107018.
[23] S.-B. Luo, J.-Y. Yu, S.-H. Yu, R. Sun, L.-Q. Cao, W.-H. Liao, C.-P. Wong, Significantly enhanced electrostatic energy storage performance of flexible polymer composites by introducing highly insulating-ferroelectric microhybrids as fillers, Adv. Energy Mater. 9 (5) (2019) 1803204.
[24] D. Yue, J.-H. Yin, W.-C. Zhang, X.-X. Cheng, M.-H. Zhang, J.-J. Wang, Y. Feng, Computational simulation for breakdown and energy storage performances with optimization in polymer dielectrics, Adv. Funct. Mater. 33 (30) (2023) 2300658.
[25] X. Zhang, Y. Shen, B. Xu, Q.-H. Zhang, L. Gu, J.-Y. Jiang, J. Ma, Y.-H. Lin, C.-W. Nan, Giant energy density and improved discharge efficiency of solution-processed polymer nanocomposites for dielectric energy Storage, Adv. Mater. 28 (10) (2016) 2055–2061.
[26] T.-D. Huan, S. Boggs, G. Teyssedre, C. Laurent, M. Cakmak, S. Kumar, R. Ramprasad, Advanced polymeric dielectrics for high energy density applications, Prog. Mater. Sci. 83 (2016) 236–269.
[27] Y. Shen, Y.-H. Lin, Q.-M. Zhang, Polymer nanocomposites with high energy storage densities, MRS Bull. 40 (9) (2015) 753–759.
[28] M.-J. Feng, Y. Feng, T.-D. Zhang, J.-L. Li, Q.-G. Chen, Q.-G. Chi, Q.-Q. Lei, Recent advances in multilayer-structure dielectrics for energy storage application, Adv. Sci. 8 (23) (2021) 2102221.
[29] B.-H. Fan, M.-Y. Zhou, C. Zhang, D.-L. He, J.-B. Bai, Polymer-based materials for achieving high energy density film capacitors, Prog. Polym. Sci. 97 (2019) 101143.
[30] Y. Guo, D. Zhou, R.-T. Li, D. Li, W.-C. Zhao, L.-X. Pang, Z.-Q. Shi, W.-F. Liu, J.-Z. Su, T. Zhou, G.-H. Chen, Novel relaxor ferroelectric BTWO nanofillers for improving the energy storage performance of polymer-based dielectric composites, J. Energy Storage 76 (2024) 109585.
[31] Y.-A. Shang, Y. Feng, C.-H. Zhang, T.-D. Zhang, Q.-Q. Lei, Q.-G. Chi, Double gradient composite dielectric with high energy density and efficiency, J. Mater. Chem. A 10 (28) (2022) 15183–15195.
[32] M.-F. Guo, J.-Y. Jiang, Z.-H. Shen, Y.-H. Lin, C.-W. Nan, Y. Shen, High-energy-density ferroelectric polymer nanocomposites for capacitive energy storage: enhanced breakdown strength and improved discharge efficiency, Mater. Today 29 (2019) 49–67.
[33] D.-X. Li, X.-J. Zeng, Z.-P. Li, Z.-Y. Shen, H. Hao, W.-Q. Luo, X.-C. Wang, F.-S. Song, Z.-M. Wang, Y.-M. Li, Progress and perspectives in dielectric energy storage ceramics, J. Adv. Ceram. 10 (4) (2021) 675–703.
[34] A. Tsyganov, M. Vikulova, D. Artyukhov, A. Bainyashev, V. Goffman, A. Gorokhovsky, E. Boychenko, I. Burmistrov, N. Gorshkov, Permittivity and Dielectric Loss Balance of PVDF/K1.6Fe1.6Ti6.4O16/MWCNT Three-Phase Composites, Polymers 14 (21) (2022) 4609.
[35] M.-H. Yang, Q.-M. Li, X.-M. Zhang, E. Bilotti, C. Zhang, C. Xu, S.-H. Gan, Z.-M. Dang, Surface engineering of 2D dielectric polymer films for scalable production of high-energy-density films, Prog. Mater. Sci. 128 (2022) 100968.
[36] X.-P. Hu, K.-W. Yi, J. Liu, B.-J. Chu, High Energy Density Dielectrics Based on PVDF-Based Polymers, Energy Technol. 6 (5) (2018) 849–864.
[37] Y.-L. Yang, J.-J. Wang, R.-Y. Jing, W.-J. Shi, L.-Y. Zhang, C. Li, X.-Y. Zeng, F.-K. Chen, G. Liu, X.-L. Chao, Y. Yan, L. Jin, Achieving high dielectric energy-storage properties through a phase coexistence design and viscous polymer process in BNT-based ceramics, J. Materiomics 9 (6) (2023) 1004–1014.
[38] C.-W. Nan, Effective-medium theory of piezoelectric composites, J. Appl. Phys. 76 (1994) 1155–1163.
[39] Y.-B. Shen, S.-B. Luo, S.-H. Yu, R. Sun, C.P. Wong, Surface-modified barium titanate by MEEAA for high-energy storage application of polymer composites, High Volt. 1 (4) (2016) 175–180.
[40] G.C. Manika, G.C. Psarras, Energy storage and harvesting in BaTiO3/epoxy nanodielectrics, High Volt. 1 (4) (2016) 151–157.
[41] Q. Wang, L. Zhu, Polymer nanocomposites for electrical energy storage, J. Polym. Sci. Pt. B-Polym. Phys. 49 (20) (2011) 1421–1429.
[42] A. Shandilya, L.S. Schadler, R. Sundararaman, First-principles identification of localized trap states in polymer nanocomposite interfaces, J. Mater. Res. 35 (8) (2020) 931–939.
[43] C. Bauer, M. Ries, S. Pfaller, Accelerating molecular dynamics simulations by a hybrid molecular dynamics-continuum mechanical approach, Soft Mater. 20 (4) (2022) 428–443.
[44] J.F. Ogilvie, Chemistry and quantum mechanics, J. Chin. Chem. Soc. 69 (1) (2022) 13–21.
[45] M.R. Tonks, L.K. Aagesen, The phase field method: mesoscale simulation aiding material discovery, Ann. Rev. Mater. Res. 49 (2019) 79–102.
[46] Z. Gong, Y.-Z. Wu, L. Wu, H. Sun, Predicting thermodynamic properties of alkanes by high-throughput force field simulation and machine learning, J. Chem Inf. Model. 58 (12) (2018) 2502–2516.
[47] Y.-Q. Gong, C.-Y. Tang, C.-P. Tsui, J.-P. Fan, Modelling of ionic polymer-metal composites by a multi-field finite element method, Int. J. Mech. Sci. 51 (11–12) (2009) 741–751.
[48] S.-L. Zhong, Z.-M. Cai, J.-W. Zha, Y. Zhao, Z.-M. Dang, Effect of interparticle electrostatic interactions on the dielectric response of 0-3 connectivity particle/polymer composites for high energy density storage, J. Appl. Phys. 127 (18) (2020) 184106.
[49] S.-L. Zhong, Z.-M. Dang, J.-W. Zha, Prediction on effective permittivity of 0-3 connectivity particle/polymer composites at low concentration with finite element method, IEEE Trns. Dielectr. Electr. Insul. 25 (6) (2018) 2122–2128.
[50] D.J. Audus, J.J. de Pablo, Polymer informatics: opportunities and challenges, ACS Macro Lett. 6 (10) (2017) 1078–1082.
[51] T. Zhou, Z. Song, K. Sundmacher, Big data creates new opportunities for materials research: a review on methods and applications of machine learning for materials design, Engineering 5 (6) (2019) 1017–1026.
[52] R.-L. Liu, J. Wang, Z.-H. Shen, Y. Shen, AI for dielectric capacitors, Energy Storage Mater. 71 (2024) 103612.
[53] R. Gurnani, S. Shukla, D. Kamal, C. Wu, J. Hao, C. Kuenneth, P. Aklujkar, A. Khomane, R. Daniels, A.A. Deshmukh, Y. Cao, G. Sotzing, R. Ramprasad, AI-assisted discovery of high-temperature dielectrics for energy storage, Nat. Commun. 15 (1) (2024) 6107.
[54] G. Pilania, C.-C. Wang, X. Jiang, S. Rajasekaran, R. Ramprasad, Accelerating materials property predictions using machine learning, Sci Rep 3 (2013) 2810.
[55] D. Li, M.-L. Li, Z.-L. Huang, Applying the properties of neurons in machine learning: a brain-like neural model with interactive stimulation for data classification, Brain Sci. 12 (9) (2022) 1191.
[56] X.-Z. Xu, T.-M. Liang, J. Zhu, D. Zheng, T.-H. Sun, Review of classical dimensionality reduction and sample selection methods for large-scale data processing, Neurocomputing 328 (2019) 5–15.
[57] J. Schmidt, M.R.G. Marques, S. Botti, M.A.L. Marques, Recent advances and applications of machine learning in solid-state materials science, NPJ Comput. Mater. 5 (2019) 83.
[58] A.-Y.-T. Wang, R.J. Murdock, S.K. Kauwe, A.O. Oliynyk, A. Gurlo, J. Brgoch, K. A. Persson, T.D. Sparks, Machine learning for materials scientists: an introductory guide toward best practices, Chem. Mat. 32 (12) (2020) 4954–4965.
[59] P.R. Kumar, B.M. Rao, C.K.R. Maddireddy, S. Thakor, C.R. Vaja, K. Prakash, P. Jain, Dielectric characterization and machine learning-based predictions in polymer composites with mixed nanoparticles, J. Macromol. Sci. Part B-Phys. (2024) 1–15.
[60] O.A. von Lilienfeld, K. Burke, Retrospective on a decade of machine learning for chemical discovery, Nat. Commun. 11 (1) (2020) 4895.
[61] J.M. Rickman, T. Lookman, S.V. Kalinin, Materials informatics: From the atomic-level to the continuum, Acta Mater. 168 473-510.
[62] V. Bhuvaneswari, M. Priyadharshini, C. Deepa, D. Balaji, L. Rajeshkumar, M. Ramesh, Deep learning for material synthesis and manufacturing systems: A review, Materials Today: Proceedings 46 (2020) 3263-3269.
[63] E.A. Olivetti, J.M. Cole, E. Kim, O. Kononova, G. Ceder, T.Y.J. Han, A. M. Hiszpanski, Data-driven materials research enabled by natural language processing and information extraction, Appl. Phys. Rev. 7 (4) (2020) 041317.
[64] J.-B. Li, Y.-Z. Peng, L.-P. Zhao, G.-D. Chen, L. Zeng, G.-Q. Wei, Y.-H. Xu, Machine-learning-assisted discovery of perovskite materials with high dielectric breakdown strength, Mater. Adv. 3 (2022) 8639–8646.
[65] C.R. Vaja, V.A. Rana, S. Thakor, P. Jain, Ac/dc conductivity and ML-based evaluation of electric characteristics of methylene blue solution, J. Mol. Liq. 410 (2024) 125676.
[66] L.-H. Chen, C. Kim, R. Batra, J.P. Lightstone, C. Wu, Z.-Z. Li, A.A. Deshmukh, Y.-F. Wang, H.-D. Tran, P. Vashishta, G.A. Sotzing, Y. Cao, R. Ramprasad, Frequency-dependent dielectric constant prediction of polymers using machine learning, NPJ Comput. Mater. 6 (1) (2020) 61.
[67] Z.-H. Shen, Z.-W. Bao, X.-X. Cheng, B.-W. Li, H.-X. Liu, Y. Shen, L.-Q. Chen, X.-G. Li, C.-W. Nan, Designing polymer nanocomposites with high energy density using machine learning, NPJ Comput. Mater. 7 (1) (2021) 110.
[68] Y.-L. Liu, C. Niu, Z. Wang, Y. Gan, Y. Zhu, S.-H. Sun, T. Shen, Machine learning in materials genome initiative: a review, J. Mater. Sci. Technol. 57 (2020) 113–122.

[69] S. Wu, Y. Kondo, M.-A. Kakimoto, B. Yang, H. Yamada, I. Kuwajima, G. Lambard, K. Hongo, Y.-B. Xu, J. Shiomi, C. Schick, J. Morikawa, R. Yoshida, Machine-learning-assisted discovery of polymers with high thermal conductivity using a molecular design algorithm, NPJ Comput. Mater. 5 (2019) 5.
[70] P. Jain, P.K. Sahoo, A.D. Khaleel, A.J.A. Al-Gburi, Enhanced prediction of metamaterial antenna parameters using advanced machine learning regression models, Progr. Electrom. Resea. C 146 (2024) 1–12.
[71] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, E. Duchesnay, Scikit-learn: Machine learning in Python, J. Mach. Learn. Res. 12 (2011) 2825–2830.
[72] J.-G. Hao, T.K. Ho, Machine learning made easy: a review of scikit-learn package in python programming language, J. Educ. Behav. Stat. 44 (3) (2019) 348–361.
[73] L. Tao, G. Chen, Y. Li, Machine learning discovery of high-temperature polymers, Patterns 2 (4) (2021) 100225.
[74] C. Kim, A. Chandrasekaran, T.D. Huan, D. Das, R. Ramprasad, Polymer genome: a data-powered polymer informatics platform for property predictions, J. Phys. Chem. C 122 (31) (2018) 17575–17585.
[75] R.-M. Ma, T.-F. Luo, PI1M: a benchmark database for polymer informatics, J. Chem Inf. Model. 60 (10) (2020) 4684–4690.
[76] A. Gopakumar, K. Pal, C. Wolverton, Identification of high-dielectric constant compounds from statistical design, npj Comput. Mater. 8 (1) (2022) 146.
[77] M.-X. Zhu, Q.-C. Yu, H.-G. Song, T.-X. Chen, J.-M. Chen, Rational design of high-energy-density polymer composites by machine learning approach, ACS Appl. Energ. Mater. 4 (2) (2021) 1449–1458.
[78] C. Yan, X.-M. Feng, C. Wick, A. Peters, G.-Q. Li, Machine learning assisted discovery of new thermoset shape memory polymers based on a small training dataset, Polymer 214 (2021) 123351.
[79] C. Kim, R. Batra, L.-H. Chen, H. Tran, R. Ramprasad, Polymer design using genetic algorithm and machine learning, Comput. Mater. Sci. 186 (2021) 110067. cc.
[80] Y. Feng, W.-X. Tang, Y. Zhang, T.-D. Zhang, Y.-N. Shang, Q.-G. Chi, Q.-G. Chen, Q.-Q. Lei, Machine learning and microstructure design of polymer nanocomposites for energy storage application, High Volt. 7 (2) (2022) 242–250.
[81] Z.-L. Liang, Z.-W. Li, S. Zhou, Y.-W. Sun, J.-Y. Yuan, C.-S. Zhang, Machine-learning exploration of polymer compatibility, Cell Rep. Phys. Sci. 3 (6) (2022) 100931.
[82] S. Curtarolo, D. Morgan, K. Persson, J. Rodgers, G. Ceder, Predicting crystal structures with data mining of quantum calculations, Phys. Rev. Lett. 91 (13) (2003) 135503.
[83] P. Jain, M.T. Islam, A.S. Alshammari, Comparative analysis of machine learning techniques for metamaterial absorber performance in terahertz applications, Alex. Eng. J. 103 (2024) 51–59.
[84] D. Yue, Y. Feng, X.-X. Liu, J.-H. Yin, W.-C. Zhang, H. Guo, B. Su, Q.-Q. Lei, Prediction of energy storage performance in polymer composites using high-throughput stochastic breakdown simulation and machine learning, Adv. Sci. 9 (17) (2022) 2105773.
[85] Z.-H. Shen, J.-J. Wang, J.-Y. Jiang, S.-X. Huang, Y.-H. Lin, C.-W. Nan, L.-Q. Chen, Y. Shen, Phase-field modeling and machine learning of electric-thermal-mechanical breakdown of polymer-based dielectrics, Nat. Commun. 10 (2019) 1843.
[86] A. Mukherjee, B.D. Ghosh, S. Roy, K.L. Goh, Ultra strong flexible $Ba_{0.7}Sr_{0.3}Zr_{0.02}Ti_{0.98}O_3$/MWCNT/PVDF Nanocomposites: Pioneering material with remarkable energy storage for Self-Powered devices, Chem. Eng. J. 488 (2024) 151014.
[87] K.N. Shah, P. Jain, S. Thakor, V.A. Rana, Machine learning analysis of low-frequency impedance spectra of binary mixtures of polar and non polar liquids, J. Macromol. Sci. Part B-Phys. (2024) 1–14.
[88] F. Thabit, O. Can, R.U.Z. Wani, M.A. Qasem, S.B. Thorat, H.A. Alkhzaimi, Data security techniques in cloud computing based on machine learning algorithms and cryptographic algorithms: Lightweight algorithms and genetics algorithms, Concurr. Comput.-Pract. Exp. 35 (21) (2023) e7691.
[89] A. Ouadah, L. Zemmouchi-Ghomari, N. Salhi, Selecting an appropriate supervised machine learning algorithm for predictive maintenance, Int. J. Adv. Manuf. Technol. 119 (7–8) (2022) 4277–4301.
[90] P. Jain, S. Thakor, A. Joshi, K.V. Chauhan, C.R. Vaja, Machine learning-driven analysis of dielectric response in polymethyl methacrylate nanocomposites reinforced with multi-walled carbon nanotubes, J. Mater. Sci.-Mater. Electron. 35 (20) (2024) 1419.
[91] S.-H. Lu, Q.-H. Zhou, Y.-X. Ouyang, Y.-L. Guo, Q. Li, J.-L. Wang, Accelerated discovery of stable lead-free hybrid organic-inorganic perovskites via machine learning, Nat. Commun. 9 (2018) 3405.
[92] P. Raccuglia, K.C. Elbert, P.D.F. Adler, C. Falk, M.B. Wenny, A. Mollo, M. Zeller, S.A. Friedler, J. Schrier, A.J. Norquist, Machine-learning-assisted materials discovery using failed experiments, Nature 533 (7601) (2016) 73–76.
[93] C. Mair, G. Kadoda, M. Lefley, K. Phalp, C. Schofield, M. Shepperd, S. Webster, An investigation of machine learning based prediction systems, J. Syst. Softw. 53 (2000) 23–29.
[94] N. Zhan, J.R. Kitchin, Uncertainty quantification in machine learning and nonlinear least squares regression models, AICHE J. 68 (6) (2022) e17516.
[95] H. Li, T.-N. Yang, Y. Zhou, D. Ai, B. Yao, Y. Liu, L. Li, L.-Q. Chen, Q. Wang, Enabling high-energy-density high-efficiency ferroelectric polymer nanocomposites with rationally designed nanofillers, Adv. Funct. Mater. 31 (1) (2021) 2006739.
[96] D. Ai, H. Li, Y. Zhou, L.-L. Ren, Z.-B. Han, B. Yao, W. Zhou, L. Zhao, J.-M. Xu, Q. Wang, Tuning nanofillers in in situ prepared polyimide nanocomposites for high-temperature capacitive energy storage, Adv. Energy Mater. 10 (16) (2020) 1903881.
[97] R.-Y. Li, Q. Deng, D. Tian, D.-Y. Zhu, B. Lin, Predicting perovskite performance with multiple machine-learning algorithms, Crystals 11 (7) (2021) 818.
[98] J. Wang, An intuitive tutorial to Gaussian processes regression, Comput. Sci. Eng. 25 (4) (2023) 4–11.
[99] V.L. Deringer, A.P. Bartók, N. Bernstein, D.M. Wilkins, M. Ceriotti, G. Csányi, Gaussian process regression for materials and molecules, Chem. Rev. 121 (16) (2021) 10073–10141.
[100] M. Schonlau, R.-Y.-Y. Zou, The random forest algorithm for statistical learning, Stata J. 20 (1) (2020) 3–29.
[101] P. Jain, A. Joshi, A. Joshi, Assessing the efficacy of machine learning models in hydroxyapatite nano-powder assisted electro discharge machining of Ti-6Al-4 V Grade-5 alloy, International Journal on Interactive Design and Manufacturing (IJIDeM) (2024:) 1–9.
[102] D.D. Lee, H.S. Seung, Learning the parts of objects by non-negative matrix factorization, Nature 401 (6755) (1999) 788–791.
[103] S.-M. Yang, Z. Yi, M. Ye, X.-F. He, Convergence analysis of graph regularized non-negative matrix factorization, IEEE Trans. Knowl. Data Eng. 26 (9) (2014) 2151–2165.
[104] B.M. Lake, R. Salakhutdinov, J.B. Tenenbaum, Human-level concept learning through probabilistic program induction, Science 350 (6266) (2015) 1332–1338.
[105] A. Krizhevsky, I. Sutskever, G.E. Hinton, Imagenet classification with deep convolutional neural networks, Commun. ACM 60 (6) (2017) 84–90.
[106] I.J. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, S. Courville, Y. Bengio, Generative adversarial nets, Advances in Neural Information Processing Systems 27 (2014) 2672–2680.
[107] Y. Bengio, A. Courville, P. Vincent, Representation learning: A review and new perspectives, IEEE Trans. Pattern Anal. Mach. Intell. 35 (8) (2013) 1798–1828.
[108] A. Kalinov, R. Bychkov, A. Ivanov, A. Osinsky, D. Yarotsky, Machine learning-assisted PAPR reduction in massive MIMO, IEEE Wirel. Commun. Lett. 10 (3) (2020) 537–541.
[109] Y. Rimal, N. Sharma, A. Alsadoon, The accuracy of machine learning models relies on hyperparameter tuning: student result classification using random forest, randomized search, grid search, bayesian, genetic, and optuna algorithms, Multimed. Tools Appl. (2024) 1–16.
[110] J. Bergstra, Y. Bengio, Random search for hyper-parameter optimization, J. Mach. Learn. Res. 13 (2) (2012) 281–305.
[111] X. He, K.-Y. Zhao, X.-W. Chu, AutoML: A survey of the state-of-the-art, Knowledge-Based Syst. 212 (2021) 106622.
[112] A.A. Chowdhury, M.A. Hossen, M.A. Azam, M.H. Rahman, Deepqgho: Quantized greedy hyperparameter optimization in deep neural networks for on-the-fly learning, IEEE Access 10 (2022) 6407–6416.